

EJERCICIOS RESUELTOS TP 12

1) a) $f(x,y) = \frac{y}{x} + \frac{y^2}{x^2}$

$f(tx,ty) = \frac{ty}{tx} + \frac{t^2 y^2}{t^2 x^2} = \frac{y}{x} + \frac{y^2}{x^2} = f(x,y)$

$z = \frac{y}{x} \Rightarrow y = zx \Rightarrow y' = z'x + z$

$z'x + z = z + z^2 \Rightarrow \frac{dz}{dx} x = z^2 \Rightarrow \int \frac{dz}{z^2} = \int \frac{dx}{x} \Rightarrow$

$\Rightarrow -\frac{1}{z} = \ln x + \ln C \Rightarrow -\frac{1}{z} = \ln Cx \Rightarrow$

$\Rightarrow -\frac{x}{y} = \ln Cx \Rightarrow e^{-x/y} = Cx$

$y(1) = 1 \Rightarrow e^{-1} = C$

$e^{-x/y} = x e^{-1}$

b) $(x^2 + y^2) dx - 2xy dy = 0 \Rightarrow (x^2 + y^2) dx = 2xy dy \Rightarrow$

$\Rightarrow \frac{dy}{dx} = \frac{x^2 + y^2}{2xy} = \frac{1}{2} \left[\frac{x}{y} + \frac{y}{x} \right] = \frac{1}{2} \left[\left(\frac{1}{z} \right)' + \frac{1}{z} \right]$

$f(tx,ty) = \frac{t^2 x^2 + t^2 y^2}{2tx \cdot ty} = \frac{t^2}{t^2} \frac{x^2 + y^2}{2xy} = f(x,y)$

$z = \frac{y}{x} \Rightarrow y = zx \Rightarrow y' = z'x + z$

$z'x + z = \frac{x^2 + z^2 x^2}{2x \cdot zx} = \frac{1 + z^2}{2z} \Rightarrow z'x = \frac{1 + z^2}{2z} - z \Rightarrow$

$\Rightarrow \frac{dz}{dx} x = \frac{1 + z^2 - 2z^2}{2z} \Rightarrow \int \frac{2z}{1 - z^2} = \int \frac{dx}{x} \Rightarrow$

$\Rightarrow \ln x + \ln k = \int \left[-\frac{1}{z} \ln(1 - z^2) \right] \Rightarrow \ln(kx) = \ln(1 - z^2)^{-1} \Rightarrow$

$\Rightarrow kx = \frac{1}{1 - z^2} \Rightarrow kx = \frac{1}{1 - \left(\frac{y}{x}\right)^2} \Rightarrow kx = \frac{x^2}{x^2 - y^2} \Rightarrow$

$\Rightarrow k(x^2 - y^2) = x$

$kx = \frac{x^2}{x^2 - y^2} \Rightarrow$

$\Rightarrow k = \frac{\frac{1}{k} x^2}{x^2 - y^2} \Rightarrow$

$\Rightarrow 0 = \frac{Cx^2}{x^2 - y^2} - k$

Ver si puedo resolverlo como DTE

$P = x^2 + y^2 \quad Q = -2xy$

$P'_y = 2y \quad Q'_x = -2y$ no es DTE

$$c) y' = \frac{y + x \cos^2(y/x)}{x}$$

$$f(tx, ty) = \frac{ty + tx \cos^2(\frac{ty}{tx})}{tx} = \frac{y + x \cos^2(\frac{y}{x})}{x} = f(x, y)$$

$$z = \frac{y}{x} \Rightarrow y = zx \Rightarrow y' = z'x + z$$

$$z'x + z = \frac{y}{x} + \cos^2\left(\frac{y}{x}\right) \Rightarrow z'x + z = \frac{y}{x} + \cos^2 z \Rightarrow$$

$$\Rightarrow \frac{dz}{dx} x = \cos^2 z \Rightarrow \int \frac{dz}{\cos^2 z} = \int \frac{dx}{x} \Rightarrow$$

$$\Rightarrow \tan z = \ln x + \ln c \Rightarrow \tan \frac{y}{x} = \ln xc \Rightarrow$$

$$\Rightarrow e^{\tan \frac{y}{x}} = cx$$

$$y(1) = \frac{\pi}{4} \Rightarrow e^{\tan \frac{\pi}{4}} = c \Rightarrow c = e$$

$$e^{\tan \frac{y}{x}} = ex \Rightarrow x = \frac{e^{\tan \frac{y}{x}}}{e} \Rightarrow \boxed{x = \frac{e^{\tan \frac{y}{x}} - 1}{y}}$$

$$d) \quad y' = \frac{y}{x-y} = \frac{\frac{y}{x}}{1 - \frac{y}{x}} = \frac{\frac{y}{x}}{1 - \frac{y}{x}}$$

$$f(tx, ty) = \frac{ty}{tx - ty} = \frac{\cancel{t}y}{\cancel{t}(x - y)} = \frac{y}{x - y}$$

$$\Rightarrow f(tx, ty) = \frac{y}{x - y} = f(x, y)$$

$$z = \frac{y}{x} \Rightarrow z x = y \Rightarrow y' = z' x + z$$

$$z' x + z = \frac{z x}{x - z x} = \frac{z x}{x(1 - z)} \quad \text{für } x > 0 \Rightarrow$$

$$z' x + z = \frac{z x}{x(1 - z)} \Rightarrow \frac{dz}{dx} x = \frac{z}{1 - z} - z \Rightarrow$$

$$\Rightarrow dz \cdot \frac{1 - z}{z - z(1 - z)} = \frac{dx}{x} \Rightarrow$$

$$\Rightarrow \int dz \cdot \frac{1 - z}{z - z + z^2} = \int \frac{dx}{x} \Rightarrow \int \frac{1}{z^2} dz - \int \frac{1}{z} dz = \ln x +$$

$$- \ln c \Rightarrow -z^{-1} - \ln z = \ln x \Rightarrow$$

$$\Rightarrow -\frac{1}{z} = \ln x + \ln z \Rightarrow -\frac{1}{\frac{y}{x}} = \ln(xz) \Rightarrow$$

$$\Rightarrow -\frac{x}{y} = \ln\left(cx \frac{y}{x}\right) \Rightarrow -\frac{x}{y} = \ln cy \Rightarrow$$

$$\Rightarrow e^{-x/y} = cy$$

Halle la familia de curvas ortogonales a $x^2+y^2=2ax$

$$2) \quad x^2+y^2=2ax \rightarrow a = \frac{x^2+y^2}{2x}$$

$$2x+2y y' = 2a \Rightarrow x+y y' = \frac{x^2+y^2}{2x}$$

$$y y' = \frac{x^2+y^2}{2x} - x = \frac{x^2+y^2-2x^2}{2x} = \frac{y^2-x^2}{2x} \Rightarrow y' = \frac{y^2-x^2}{2xy}$$

ED del haz
haz
perpend. $\frac{-1}{y} = \frac{y^2-x^2}{2xy} \rightarrow y' = \frac{-2xy}{y^2-x^2}$ Verifico si la ED es homogénea de 1er orden

$$f(x,y) = \frac{-2xy}{y^2-x^2}$$

$$f(tx, ty) = \frac{-2txty}{t^2y^2-t^2x^2} = \frac{-2xy}{y^2-x^2} = f(x,y)$$

$$z = \frac{y}{x} \Rightarrow y = zx \Rightarrow y' = z'x + z$$

$$y' = \frac{-2xy}{y^2-x^2} \Rightarrow z'x + z = \frac{-2xzx}{z^2x^2-x^2} \Rightarrow$$

$$\Rightarrow z'x = \frac{-2z}{z^2-1} - z \Rightarrow \frac{dz}{dx} x = \frac{-2z - z(z^2-1)}{z^2-1} \Rightarrow$$

$$\Rightarrow \int dz \frac{z^2-1}{-2z-z^3+z} = \int \frac{dx}{x} \Rightarrow \int \frac{z^2-1}{-2z-z^3+z} = \ln kx \Rightarrow$$

$$\Rightarrow \int dz \frac{1-z^2}{z+z^3} = \ln kx \Rightarrow \int \frac{dz}{z(1+z^2)} - \int \frac{z}{1+z^2} dz = \ln kx \Rightarrow$$

$$\Rightarrow \frac{1}{2} \ln \frac{z^2}{1+z^2} - \frac{1}{2} \ln (1+z^2) = \ln kx \Rightarrow$$

$$\Rightarrow \frac{1}{2} \left(\ln \left[\frac{z^2}{1+z^2} \cdot (1+z^2) \right] \right) = \ln kx \Rightarrow$$

$$\Rightarrow \frac{1}{2} \ln \left[\frac{z^2}{(1+z^2)^2} \right] = \ln kx \Rightarrow$$

$$\Rightarrow \ln \sqrt{\left(\frac{z}{1+z^2} \right)^2} = \ln kx \Rightarrow \frac{z}{1+z^2} = kx \Rightarrow$$

$$\Rightarrow \frac{\frac{y}{x}}{1+\left(\frac{y}{x}\right)^2} = kx \Rightarrow \frac{\frac{y}{x}}{\frac{x^2+y^2}{x^2}} = kx \Rightarrow \frac{y}{x^2+y^2} = kx$$

$$\Rightarrow \frac{y}{k} = x^2+y^2 \Rightarrow \boxed{x^2+y^2 = cy}$$

NOTA

3))

$$x = u + v$$

$$y = r + k \Rightarrow y' = r'$$

$$r' = \frac{u+v - r-k-1}{u+v+r+k+3} = \frac{u-r}{u+r} \Leftrightarrow \begin{cases} h-k-1=0 \\ h+k+3=0 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} h=k+1 \\ k+1+k+3=0 \end{cases} \Rightarrow \begin{cases} h=k+1 \\ 2k=-4 \end{cases} \Rightarrow \begin{cases} k=-2 \\ h=-2+1 \end{cases} \Rightarrow \begin{cases} k=-2 \\ h=-1 \end{cases}$$

$$\begin{cases} x = u-1 \\ y = r-2 \end{cases} \Rightarrow \frac{dr}{du} = \frac{u-r}{u+r}$$

$$f(tu, tr) = \frac{tu - tr}{tu + tr} = \frac{u - r}{u + r} = f(u, r)$$

$$z = \frac{r}{u} \Rightarrow uz = r \Rightarrow z'u + z = r'$$

$$z'u + z = \frac{u - uz}{u + uz} \Rightarrow \frac{dz}{du} u = \frac{u(1-z)}{u(1+z)} - z = 0$$

$$\Rightarrow \frac{dz}{du} u = \frac{1-z-z(1+z)}{1+z} = \frac{1-z-z-z^2}{1+z} = 0$$

$$\Rightarrow \frac{dz}{du} u = \frac{1-2z-z^2}{1+z} \Rightarrow \int dz \frac{1+z}{1-2z-z^2} = \int \frac{du}{u} \Rightarrow$$

$$\Rightarrow \int \frac{dz}{-z^2-2z+1} + \int \frac{dz \cdot z}{-z^2-2z+1} = \ln u + \ln c = 0 \quad (1)$$

$$a = -1, b = -2, c = 1 \quad \Delta = 4ac - b^2 = 4(-1) \cdot 1 - 4 = -8$$

$$(1) \Rightarrow +\frac{1}{\sqrt{8}} \ln \frac{-2z-2-\sqrt{8}}{-2z-2+\sqrt{8}} + \frac{1}{-2} \ln(-z^2-2z+1) - \frac{(-2)}{2(-1)} =$$

$$\cdot \left(+\frac{1}{\sqrt{8}} \ln \frac{-2z-2-\sqrt{8}}{-2z-2+\sqrt{8}} \right) = \ln cu = 0$$

$$\Rightarrow \ln \left(\frac{-2z-2-\sqrt{8}}{-2z-2+\sqrt{8}} \right)^{1/\sqrt{8}} + \ln(-z^2-2z+1)^{-1/2} =$$

$$= \ln \left(\frac{-2z-2-\sqrt{8}}{-2z-2+\sqrt{8}} \right)^{1/\sqrt{8}} = \ln cu = 0$$

$$\Rightarrow \ln \left[\left(\frac{-2z-2-\sqrt{8}}{-2z-2+\sqrt{8}} \right)^{1/\sqrt{8}} / \left(\frac{-2z-2-\sqrt{8}}{-2z-2+\sqrt{8}} \right)^{1/\sqrt{8}} \right] (-z^2-2z+1)^{-1/2} =$$

$$= \ln cu = 0$$

$$\Rightarrow \ln(-z^2-2z+1)^{-1/2} = \ln cu \Rightarrow (-z^2-2z+1)^{-1/2} = cu \Rightarrow$$

$$\Rightarrow \left[-\left(\frac{r}{u}\right)^2 - 2\left(\frac{r}{u}\right) + 1 \right]^{-1/2} = cu \Rightarrow \left[-\frac{r^2}{u^2} - \frac{2r}{u} + 1 \right]^{-1/2} = cu \Rightarrow$$

$$\Rightarrow \left[\frac{-r^2-2ur+u^2}{u^2} \right]^{-1/2} = cu \Rightarrow \frac{\sqrt{u^2}}{\sqrt{-r^2-2ur+u^2}} = cu \Rightarrow$$

$$\Rightarrow 1 = c \sqrt{-r^2-2ur+u^2} \quad .$$

$$\Rightarrow 1 = \frac{c^2}{c^2} (-r^2 - 2ur + u^2) \Rightarrow \frac{1}{c^2} = - (y+2)^2 -$$

$$- 2(x+1)(y+2) + (x+1)^2 \Rightarrow$$

$$\Rightarrow k = - (y+2)^2 - 2(x+1)(y+2) + (x+1)^2$$

$$k = -y^2 - 4y - 4 - 2xy - 4x - 2y - 4 + x^2 + 2x + 1$$

4)

$$\begin{aligned} a) \quad P &= 2xy \\ Q &= x^2 + \cos y \end{aligned}$$

$$\begin{aligned} P'_y &= 2x \\ Q'_x &= 2x \end{aligned} \quad \left. \vphantom{\begin{aligned} P'_y &= 2x \\ Q'_x &= 2x \end{aligned}} \right\} \rightarrow DTE$$

Por lo tanto: $\begin{cases} \varphi'_x = 2xy \\ \varphi'_y = x^2 + \cos y \end{cases}$

$$\varphi = \int 2xy \, dx = x^2 y + g(y)$$

$$\begin{aligned} \varphi'_y &= x^2 + g'(y) \\ \varphi'_y &= x^2 + \cos y \end{aligned} \quad \left. \vphantom{\begin{aligned} \varphi'_y &= x^2 + g'(y) \\ \varphi'_y &= x^2 + \cos y \end{aligned}} \right\} \Rightarrow g'(y) = \cos y = 0 \quad \int dg = \int \cos y \, dy = 0$$

$$\Rightarrow g = \sin y + C$$

La SG de la ED es de ecuación $\varphi(x,y) = \text{cte.}$

$$\varphi = x^2 y + \sin y + C = C_1 \Rightarrow \boxed{x^2 y + \sin y + k = 0}$$

$$b) \quad \frac{dy}{dx} = \frac{xy^2 - 1}{1 - x^2 y} \Rightarrow dy(1 - x^2 y) = dx(xy^2 - 1) \Rightarrow$$

$$\Rightarrow dx(xy^2 - 1) - dy(1 - x^2 y) = 0 \Rightarrow$$

$$\Rightarrow dx(xy^2 - 1) + dy(x^2 y - 1) = 0$$

$$\begin{aligned} P &= xy^2 - 1 & P'_y &= 2xy \\ Q &= x^2 y - 1 & Q'_x &= 2xy \end{aligned} \quad \left. \vphantom{\begin{aligned} P &= xy^2 - 1 \\ Q &= x^2 y - 1 \end{aligned}} \right\} DTE$$

$$\varphi = \int (xy^2 - 1) \, dx = \frac{x^2 y^2}{2} - x + g(y)$$

$$\begin{aligned} \varphi'_y &= x^2 y + g'(y) \\ \varphi'_y &= x^2 y - 1 \end{aligned} \quad \left. \vphantom{\begin{aligned} \varphi'_y &= x^2 y + g'(y) \\ \varphi'_y &= x^2 y - 1 \end{aligned}} \right\} \Rightarrow g'(y) = -1 \Rightarrow \int dg = \int -dy = 0$$

$$\Rightarrow g = -y + C$$

$$\varphi = \frac{x^2 y^2}{2} - x - y + C = C_1 \Rightarrow k = x + y - \frac{x^2 y^2}{2} = 0$$

$$y(-1) = 1 \Rightarrow k = -1 + 1 - \frac{1}{2} = 0 \Rightarrow k = -\frac{1}{2}$$

$$x+y - \frac{x^2 y^2}{2} = -\frac{1}{2} \Rightarrow x+y - \frac{x^2 y^2}{2} + \frac{1}{2} = 0$$

$$\begin{aligned} c) \quad P &= 6xy - y^3 & P'_y &= 6x - 3y^2 \\ Q &= 4y + 3x^2 - 3xy^2 & Q'_x &= 6x - 3y^2 \end{aligned} \quad \left. \vphantom{\begin{aligned} P &= 6xy - y^3 \\ Q &= 4y + 3x^2 - 3xy^2 \end{aligned}} \right\} \Rightarrow \text{DfE}$$

$$\varphi = \int (6xy - y^3) dx = 3x^2 y - y^3 x + g(y)$$

$$\left. \begin{aligned} \varphi'_y &= 3x^2 - 3y^2 x + g'(y) \\ \varphi'_y &= 4y + 3x^2 - 3xy^2 \end{aligned} \right\} \Rightarrow g'(y) = 4y \Rightarrow \int dg = \int 4y dy \Rightarrow$$

$$\Rightarrow g = 2y^2 + c$$

$$\varphi = 3x^2 y - y^3 x + 2y^2 + c = c_1 \Rightarrow \boxed{3x^2 y - y^3 x + 2y^2 + c = 0}$$

$$d) (y^2 - y) dx + x dy = 0$$

$$\begin{aligned} M &= y^2 - y & M'_x &= 2y - 1 \\ N &= x & N'_x &= 1 \end{aligned} \quad \left. \vphantom{\begin{aligned} M &= y^2 - y \\ N &= x \end{aligned}} \right\} \Rightarrow \text{no DfE}$$

$$\frac{M'_y + N'_x}{M} = f(y)$$

$$g(y) (y^2 - y) dx + g(y) x dy = 0$$

$$\begin{aligned} P &= g(y^2 - y) & P'_y &= g'(y^2 - y) + (2y - 1)g \\ Q &= g x & Q'_x &= g \end{aligned} \quad \left. \vphantom{\begin{aligned} P &= g(y^2 - y) \\ Q &= g x \end{aligned}} \right\} \Rightarrow$$

$$\Rightarrow g'(y^2 - y) + (2y - 1)g = g \Rightarrow g'(y^2 - y) = g(1 - 2y + 1) = 0$$

$$\Rightarrow \frac{dg}{dy} = \frac{2 - 2y}{y^2 - y} g \Rightarrow \int \frac{dg}{g} = \int \frac{2 - 2y}{y^2 - y} dy \Rightarrow$$

$$\Rightarrow \ln g = 2 \int \frac{dy}{y^2 - y} = 2 \int \frac{dy}{y(1+y)} \Rightarrow$$

$$\Rightarrow \ln g = 2 \left[\frac{1}{1} \ln \frac{2y-1-1}{2y-1+1} \right] + 2 \ln(y-1) \Rightarrow$$

auxiliäre

$$a=1$$

$$b=-1$$

$$c=0$$

$$\Delta = 4ac - b^2 = -1$$

$$\Rightarrow \ln g = 2 \ln \frac{2y-2}{2y} + \ln(y-1)^2 \Rightarrow$$

$$\Rightarrow \ln g = \ln \frac{(y-1)^2}{y^2} + \ln(y-1)^2 \Rightarrow$$

$$\Rightarrow \ln g = \ln \frac{(y-1)^2}{y^2} \cdot (y-1)^2 \Rightarrow g = \frac{1}{y^2}$$

$$P = \frac{1}{y^2} (y^2 - y) = 1 - \frac{1}{y}$$

$$Q = \frac{1}{y^2} \cdot x = \frac{x}{y^2}$$

$$P'_y = \frac{1}{y^2}$$

$$Q'_x = \frac{1}{y^2}$$

$\Rightarrow \text{DTE}$

$$\phi = \int Q dy \Rightarrow \phi = \int \frac{x}{y^2} dy \Rightarrow \phi = x \left(-\frac{1}{y} \right) + h(x) \Rightarrow$$

$$\Rightarrow \phi = -\frac{x}{y} + h(x)$$

$$\phi'_x = -\frac{1}{y} + h'(x) = 1 - \frac{1}{y} \Rightarrow h'(x) = 1 \Rightarrow h = \int dx \Rightarrow$$

$$\Rightarrow h = x + k$$

$$\phi = -\frac{x}{y} + x + k$$

$$\phi = C \Rightarrow -\frac{x}{y} + x + k = C \Rightarrow -\frac{x}{y} + x = C_1 \Rightarrow$$

$$\Rightarrow -C_1 + x = \frac{x}{y} \Rightarrow y = \frac{x}{x-C_1} \Rightarrow y = \frac{x}{x+C_2}$$

4) d) non sep

$$(y^2 - y) dx + x dy = 0 \Rightarrow (y^2 - y) dx = -x dy \Rightarrow$$

$$\Rightarrow \int \frac{dx}{x} = - \int \frac{dy}{y^2 - y} \Rightarrow \ln x + \ln C = - \frac{1}{1} \ln \frac{2y-1-1}{2y-1+1} \Rightarrow$$

$$\begin{aligned} a &= 1 \\ b &= -1 \\ c &= 0 \end{aligned}$$

$$\Delta = 4ac - b^2 = 0 - 1 = -1$$

$$\Rightarrow \ln Cx = - \ln \frac{2y-2}{2y} \Rightarrow Cx = \left(\frac{y-1}{y} \right)^{-1} \Rightarrow$$

$$\Rightarrow Cx = \frac{y}{y-1} \Rightarrow Cx(y-1) = y \Rightarrow Cxy - Cx = y \Rightarrow Cxy - y = Cx \Rightarrow$$

$$\Rightarrow y(Cx-1) = Cx \Rightarrow y = \frac{Cx}{Cx-1} \Rightarrow y = \frac{x}{x-\frac{1}{C}} \Rightarrow y = \frac{x}{x+k} \quad (const = -\frac{1}{C})$$

$$e) (x+xy^2) dx - 2yx dy = 0$$

$$\begin{aligned} M &= x+xy^2 & M'_y &= 2y \\ N &= -2yx & N'_x &= -2y \end{aligned} \quad \left. \vphantom{\begin{aligned} M &= x+xy^2 \\ N &= -2yx \end{aligned}} \right\} \text{no } \Rightarrow \text{BTE}$$

$$\frac{M'_y - N'_x}{N} = \frac{2y + 2y}{-2yx} = \frac{4}{-2x} = f(x)$$

$$g(x) (x+xy^2) dx - 2yx g(x) = 0$$

$$\begin{aligned} P &= g(x+xy^2) & P'_y &= g \cdot 2y \\ Q &= -2yxg & Q'_x &= -2yg + g'(-2yx) \end{aligned} \quad \left. \vphantom{\begin{aligned} P &= g(x+xy^2) \\ Q &= -2yxg \end{aligned}} \right\} = 0$$

$$\Rightarrow 2gy = -2y(g + g'x) = 0 \quad g = -g - \frac{dg}{dx} x = 0$$

$$\Rightarrow 2g = -\frac{dg}{dx} x \Rightarrow \int \frac{2}{x} dx = \int -\frac{dg}{g} \Rightarrow 2 \ln x = -\ln g \Rightarrow$$

$$\Rightarrow \ln x^2 = \ln \frac{1}{g} \Rightarrow x^2 = \frac{1}{g} \Rightarrow g = \frac{1}{x^2}$$

$$P = \frac{1}{x^2} (x+xy^2) = \frac{x+xy^2}{x^2} \Rightarrow P'_y = \frac{2y}{x^2} \quad \left. \vphantom{P = \frac{x+xy^2}{x^2}} \right\} \Rightarrow \text{BTE}$$

$$Q = \frac{1}{x^2} (-2yx) = \frac{-2y}{x} \Rightarrow Q'_x = -2y \left(-\frac{1}{x^2} \right) = \frac{2y}{x^2}$$

$$\frac{x+xy^2}{x^2} dx - \frac{2y}{x} dy = 0$$

$$\varphi = \int -\frac{2y}{x} dy = -\frac{y^2}{x} + h(x)$$

$$\varphi'_x = -y^2 \left(-\frac{1}{x^2} \right) + h'(x) = \frac{y^2}{x^2} + h'(x)$$

$$\varphi'_x = \frac{x}{x^2} + \frac{y^2}{x^2}$$

$$\Rightarrow h'(x) = \frac{1}{x} \Rightarrow$$

$$\Rightarrow h = \int \frac{1}{x} dx \Rightarrow h = \ln x + C$$

$$\varphi = -\frac{y^2}{x} + \ln x + C = C_1 \Rightarrow$$

$$\Rightarrow \boxed{\ln x - \frac{y^2}{x} + k = 0} \Rightarrow \ln x = \frac{y^2}{x} + C_2$$

$$f) y^2 \cos x \, dx + (4 + 5y \sin x) \, dy = 0$$

$$M = y^2 \cos x$$

$$M'_y = 2y \cos x$$

$$N = 4 + 5y \sin x$$

$$N'_x = 5y \cos x$$

} no $\Rightarrow$ DTE

$$\frac{M'_y - N'_x}{M} = \frac{2y \cos x + 5y \cos x}{y^2 \cos x} = \frac{-3y}{y^2} = f(y)$$

$$P = g(y) y^2 \cos x$$

$$P'_y = g' y^2 \cos x + 2y g \cos x$$

$$Q = g(y) (4 + 5y \sin x)$$

$$Q'_x = -5y g \cos x$$

} = 0

$$\Rightarrow g' y^2 \cos x + 2y g \cos x = -5y g \cos x \Rightarrow$$

$$\Rightarrow \frac{dg}{dy} y^2 = -3y g \Rightarrow \int \frac{dg}{g} = \int \frac{-3 dy}{y} \Rightarrow$$

$$\Rightarrow \ln g = -3 \ln y \Rightarrow g = y^{-3}$$

$$P = y^3 y^2 \cos x = y^5 \cos x$$

$$P'_y = 5y^4 \cos x$$

} $\Rightarrow$ DTE

$$Q = y^3 (4 + 5y \sin x) = 4y^3 + 5y^4 \sin x \quad Q'_x = 5y^4 \cos x$$

$$\varphi = \int 5y^4 \cos x \, dx = y^5 \sin x + h(y)$$

$$\varphi'_y = 5y^4 \sin x + h'(y)$$

$$\varphi'_y = 4y^3 + 5y^4 \sin x$$

$$\Rightarrow h'(y) = 4y^3 \Rightarrow h = \int 4y^3 \, dy \Rightarrow$$

$$\Rightarrow h = y^4 + c$$

$$\varphi = y^5 \sin x + y^4 + c = c_1 \Rightarrow y^5 \sin x + y^4 + k = 0$$

$$5) x^2 y' = y + x e^{-1/x}$$

hacemos $y = x^2 q(x)$

$$y' = 2x q(x) + q'(x) \cdot x^2$$

$$x^2 (2x q + q' x^2) = x^2 q + x e^{-1/x}$$

$$2x^2 q + q' x^3 = x q + e^{-1/x}$$

$$q' x^3 + q (2x^2 - x) = e^{-1/x}$$

$$q' + q \frac{2x^2 - x}{x^3} = \frac{e^{-1/x}}{x^3} \quad \text{lineal}$$

$$q = u r \Rightarrow q' = u' r + r' u$$

$$u' r + r' u + u r \frac{2x^2 - x}{x^3} = \frac{e^{-1/x}}{x^3}$$

$$u' r + u \left(r' + r \frac{2x^2 - x}{x^3} \right) = \frac{e^{-1/x}}{x^3} \quad (*)$$

$$\frac{dr}{dx} = -r \frac{2x^2 - x}{x^3} \Rightarrow \int \frac{dr}{r} = \int dx \frac{x - 2x^2}{x^3} \Rightarrow$$

$$\Rightarrow \ln r = \int \frac{1 - 2x}{x^2} dx = \int \frac{1}{x^2} dx - \int \frac{2}{x} dx = 0$$

$$\Rightarrow \ln r = -\frac{1}{x} - 2 \ln x \Rightarrow r = e^{-1/x - 2 \ln x}$$

Reemplazamos en (*) $\rightarrow u' e^{-1/x - 2 \ln x} = \frac{e^{-1/x}}{x^3} \Rightarrow$

$$\Rightarrow \frac{du}{dx} = \frac{e^{-1/x}}{x^3} : \frac{e^{-1/x}}{e^{2 \ln x}} \Rightarrow \frac{du}{dx} = \frac{e^{-1/x}}{x^3} \cdot \frac{x^2}{e^{-1/x}} \Rightarrow$$

$$\Rightarrow \frac{du}{dx} = \int \frac{1}{x^3} dx \Rightarrow u = f(x)$$

$$q = u r \Rightarrow q = \ln(kx) \frac{e^{-1/x}}{e^{2 \ln x}} \Rightarrow q = \frac{\ln(kx) e^{-1/x}}{x^2}$$

$$y = x^2 q \Rightarrow y = \ln(kx) e^{-1/x}$$

$$6)a) y'' + 8y' + 25y = 0$$

$$r^2 + 8r + 25 = 0 \Rightarrow r = \frac{-8 \pm \sqrt{64 - 100}}{2} = \frac{-8 \pm \sqrt{-36}}{2} \Rightarrow$$

$$\Rightarrow r = \frac{-8 \pm 6i}{2} \begin{matrix} \nearrow -4 + 3i \\ \searrow -4 - 3i \end{matrix}$$

$$y = e^{-4x} (A \sin(3x) + B \cos(3x))$$

$$6)b) y''' - 5y'' + 8y' - 4y = 2$$

$$y = y_H + y_P$$

$$y_H \mid r^3 - 5r^2 + 8r - 4 = 0 \quad r = 1 \text{ es raíz}$$

$$\begin{array}{r|rrrr} 1 & 1 & -5 & 8 & -4 \\ & & 1 & -4 & 4 \\ \hline & 1 & -4 & 4 & 0 \end{array}$$

$$r^2 - 4r + 4 = (r - 2)^2$$

$$\text{Luego: } r^3 - 5r^2 + 8r - 4 = (r - 1)(r - 2)^2$$

$$y_H = C_1 e^x + C_2 e^{2x} + C_3 x e^{2x}$$

$$y_P \mid y = A$$

$$y' = 0 \quad y'' = 0 \quad y''' = 0$$

$$\text{Luego: } 0 - 5 \cdot 0 + 8 \cdot 0 - 4A = 2 \Rightarrow A = -\frac{2}{4} = -\frac{1}{2}$$

$$y_P = -\frac{1}{2}$$

$$y = C_1 e^x + C_2 e^{2x} + C_3 x e^{2x} - \frac{1}{2}$$

$$6)c) \quad y'' - 3y' + 2y = xe^x + 2x \quad y_0 = y_H + y_P$$

$$\boxed{y_H} \quad \lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda = \frac{3 \pm \sqrt{9-8}}{2} = \frac{3 \pm 1}{2} \begin{matrix} 2 \\ 1 \end{matrix}$$

$$y_H = C_1 e^{2x} + C_2 e^x$$

$$c) \quad y_H = C_1 e^{2x} + C_2 e^x$$

$$y_P = u e^{2x} + v e^x$$

$$W = \begin{vmatrix} e^{2x} & e^x \\ 2e^{2x} & e^x \end{vmatrix} = e^{2x}e^x - 2e^{2x}e^x = -e^{3x}$$

$$\begin{cases} u' e^{2x} + v' e^x = 0 \\ 2u' e^{2x} + v' e^x = xe^x + 2x \end{cases}$$

$$u' = \frac{\begin{vmatrix} 0 & e^x \\ xe^x + 2x & e^x \end{vmatrix}}{-e^{3x}} = \frac{(xe^{2x} + 2xe^x)}{-e^{3x}} = -xe^{-x} - 2xe^{-2x}$$

$$u' = -xe^{-x} - 2xe^{-2x} \Rightarrow \int du = \int (-xe^{-x} - 2xe^{-2x}) dx \Rightarrow$$

$$\Rightarrow u = -\left[+e^{-x}(-x-1) + \cancel{2} \frac{e^{-2x}}{y_2}(-2x-1) \right] = 0$$

$$\Rightarrow u = -xe^{-x} - e^{-x} - xe^{-2x} - \frac{e^{-2x}}{2}$$

$$v' = \frac{\begin{vmatrix} e^{2x} & 0 \\ 2e^{2x} & xe^x + 2x \end{vmatrix}}{-e^{3x}} = \frac{xe^{3x} + 2xe^{2x}}{-e^{3x}} = -x - 2xe^{-x}$$

$$\frac{dr}{dx} = -x - 2xe^{-x} \Rightarrow \int dr = \int (-x - 2xe^{-x}) dx \Rightarrow$$

$$\Rightarrow r = -\frac{x^2}{2} - 2(e^{-x}(-x-1)) = -\frac{x^2}{2} + 2xe^{-x} + 2e^{-x}$$

$$y_p = \left(-xe^{-x} - e^{-x} - xe^{-2x} - \frac{e^{-2x}}{2} \right) e^{2x} +$$

$$+ \left(\frac{x^2}{2} + 2xe^{-x} + 2e^{-x} \right) e^x \Rightarrow$$

$$\Rightarrow y_p = -xe^x - e^x - x - \frac{1}{2} - \frac{x^2 e^x}{2} + 2x + 2 \Rightarrow$$

$$\Rightarrow y_p = -xe^x - e^x - \frac{x^2}{2} e^x + \frac{3}{2} + x$$

$$y = c_1 e^{2x} + c_2 e^x - xe^x - e^x - \frac{x^2}{2} e^x + \frac{3}{2} + x$$

$$d) \quad y'' + y = \sec x \quad \underline{y_H} \quad k^2 + 1 = 0 \rightarrow k_1 = i = 0 + i \quad \checkmark \quad k_2 = -i = 0 - i$$

$$y_H = e^{0x} (A \cos x + B \sin x)$$

$$y_H = A \cos x + B \sin x$$

$$\underline{y_P} \quad y_P = u \cos x + v \sin x$$

$$\begin{cases} u' \cos x + v' \sin x = 0 \\ -u' \sin x + v' \cos x = \sec x \end{cases}$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$u' = \frac{\begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix}}{1} = -\sin x \sec x = -\frac{\sin x}{\cos x} = -\tan x$$

$$\frac{du}{dx} = -\tan x \Rightarrow \int du = \int -\tan x \, dx \Rightarrow u = \ln |\cos x|$$

$$v' = \frac{\begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix}}{1} = \cos x \sec x = \frac{\cos x}{\cos x} = 1$$

$$\frac{dv}{dx} = 1 \Rightarrow \int dv = \int dx \Rightarrow v = x$$

$$y_P = \ln |\cos x| \cdot \cos x + x \sin x$$

$$y = A \cos x + B \sin x + \cos x \ln |\cos x| + x \sin x$$

$$e) \quad y'' - 2y' + y = \frac{e^x}{x} \quad y = y_H + y_P$$

$$\text{I.H.} \quad y'' - 2y' + y = 0$$

$$\lambda^2 - 2\lambda + 1 = 0 \Rightarrow (\lambda - 1)^2 = 0 \Rightarrow \lambda_1 = \lambda_2 = 1$$

$$y_H = C_1 e^x + C_2 x e^x$$

$$\begin{cases} u' e^x + v' x e^x = 0 \\ u' e^x + v' (e^x + x e^x) = \frac{e^x}{x} \end{cases}$$

$$W = \begin{vmatrix} e^x & x e^x \\ e^x & e^x + x e^x \end{vmatrix} = e^{2x} + \cancel{x e^{2x}} - \cancel{x e^{2x}} = e^{2x}$$

$$u' = \frac{\begin{vmatrix} 0 & x e^x \\ \frac{e^x}{x} & e^x + x e^x \end{vmatrix}}{e^{2x}} \Rightarrow u' = \frac{-e^{2x}}{e^{2x}} = -1$$

$$\frac{du}{dx} = -1 \Rightarrow \int du = \int -1 dx \Rightarrow u = -x$$

$$v' = \frac{\begin{vmatrix} e^x & 0 \\ e^x & \frac{e^x}{x} \end{vmatrix}}{e^{2x}} = \frac{\frac{e^{2x}}{x}}{e^{2x}} = \frac{1}{x}$$

$$\frac{dv}{dx} = \frac{1}{x} \Rightarrow \int dv = \int \frac{dx}{x} \Rightarrow v = \ln|x|$$

$$y_P = -x e^x + \ln|x| x e^x$$

$$y = C_1 e^x + C_2 x e^x - x e^x + x \ln|x| e^x$$

$$6) f) \quad y'' + y' = 2 \sin x$$

$$y_6 = y_H + y_P$$

$$y_H \quad \lambda^2 + \lambda = 0 \Rightarrow \lambda(\lambda + 1) = 0 \Rightarrow \lambda = 0 \vee \lambda = -1$$

$$y_H = c_1 e^{0 \cdot x} + c_2 e^{-x} = c_1 + c_2 e^{-x}$$

$$y_P \quad \text{propongo} \quad y_P = A \sin x + B \cos x$$

$$y_P' = A \cos x - B \sin x$$

$$y_P'' = -A \sin x - B \cos x$$

$$\begin{aligned} y_P'' + y_P' &= -A \sin x - B \cos x + A \cos x - B \sin x = \\ &= \sin x (-A - B) + \cos x (-B + A) \end{aligned}$$

$$\begin{cases} -A - B = 2 \\ -B + A = 0 \end{cases} \Rightarrow \begin{cases} A = B \\ -2A = 2 \end{cases} \Rightarrow A = B = -1$$

$$y_P = -\sin x - \cos x$$

$$y_6 = c_1 + c_2 e^{-x} - \sin x - \cos x$$

$$7) \frac{\partial (y g'(x))}{\partial y} = \frac{\partial (x^2 - g'(x))}{\partial x}$$

$$g'(x) + 0 \cdot y' = 2x - g''(x) \Rightarrow g'' + g' = 2x$$

$$J_H \quad r^2 + r = 0 \Rightarrow r(r+1) = 0 \Rightarrow r = 0 \vee r = -1$$

$$y_H = c_1 e^{0x} + c_2 e^{-x} \Rightarrow y_H = c_1 + c_2 e^{-x}$$

$$J_P \quad y_P = Ax^2 + Bx + C \quad y_P' = 2Ax + B \quad y_P'' = 2A$$

$$y'' + y' = 2x \Rightarrow 2A + 2Ax + B = 2x \Rightarrow$$

$$\Rightarrow \begin{cases} 2A = 2 \\ 2A + B = 0 \end{cases} \Rightarrow \begin{cases} A = 1 \\ B = -2 \end{cases} \quad \text{luego } C = 0$$

$$y_P = x^2 - 2x$$

$$y_G = c_1 + c_2 e^{-x} + x^2 - 2x$$

$$g(x) = c_1 + c_2 e^{-x} + x^2 - 2x$$

$$8) a) \quad y = y_H + y_P \quad y'' - y' - 2y = 4x^2$$

y_H

$$y'' - y' - 2y = 0$$

$$r^2 - r - 2 = 0 \Rightarrow r = \frac{1 \pm \sqrt{1+8}}{2} = \frac{1 \pm 3}{2} \begin{matrix} 2 \\ -1 \end{matrix}$$

$$y_H = c_1 e^{2x} + c_2 e^{-x}$$

y_P $y_P = u e^{2x} + v e^{-x}$

$$\begin{cases} u' e^{2x} + v' e^{-x} = 0 \\ u' 2e^{2x} - v' e^{-x} = 4x^2 \end{cases}$$

$$W = \begin{vmatrix} e^{2x} & e^{-x} \\ 2e^{2x} & -e^{-x} \end{vmatrix} = -e^x - 2e^x = -3e^x$$

$$u' = \frac{\begin{vmatrix} 0 & e^{-x} \\ 4x^2 & -e^{-x} \end{vmatrix}}{-3e^x} = \frac{-4x^2 e^{-x}}{-3e^x} = \frac{4}{3} x^2 e^{-2x}$$

$$\frac{du}{dx} = \frac{4}{3} x^2 e^{-2x} \Rightarrow \int du = \int \frac{4}{3} x^2 e^{-2x} dx \Rightarrow$$

$$\Rightarrow u = \frac{4}{3} e^{-2x} \left(\frac{x^2}{-2} - \frac{2x}{4} - \frac{2}{8} \right) \Rightarrow$$

$$\Rightarrow u = \frac{4}{3} e^{-2x} \left(-\frac{x^2}{2} - \frac{x}{2} - \frac{1}{4} \right) = \frac{e^{-2x}}{3} (-2x^2 - 2x - 1)$$

$$v' = \frac{\begin{vmatrix} e^{2x} & 0 \\ 2e^{2x} & 4x^2 \end{vmatrix}}{-3e^x} = \frac{4e^{2x} x^2}{-3e^x} = -\frac{4}{3} x^2 e^x$$

$$\frac{dv}{dx} = -\frac{4}{3} x^2 e^x \Rightarrow \int dv = \int -\frac{4}{3} x^2 e^x dx \Rightarrow$$

$$\Rightarrow v = -\frac{4}{3} e^x (x^2 - 2x + 2)$$

$$y_P = \frac{e^{-2x}}{3} e^{2x} (-2x^2 - 2x - 1) - \frac{4}{3} e^x e^{-x} (x^2 - 2x + 2)$$

$$y_p = -\frac{2}{3}x^2 - \frac{2}{3}x - \frac{1}{3} - \frac{4}{3}x^2 + \frac{8}{3}x - \frac{8}{3}$$

$$y_p = -2x^2 + 2x - 3$$

$$y = c_1 e^{2x} + c_2 e^{-x} - 2x^2 + 2x - 3$$

$$y(0) = 1 \quad y'(0) = 4$$

$$y(0) = 1 \Rightarrow 1 = c_1 + c_2 - 3$$

$$y' = 2c_1 e^{2x} - c_2 e^{-x} - 4x + 2$$

$$y'(0) = 4 \Rightarrow 4 = 2c_1 - c_2 + 2$$

$$\left. \begin{array}{l} c_1 + c_2 = 4 \\ 2c_1 - c_2 = 2 \end{array} \right\} \Rightarrow \text{MAH} \quad 3c_1 = 6 \Rightarrow c_1 = 2$$

$$c_1 + c_2 = 4 \Rightarrow 2 + c_2 = 4 \Rightarrow c_2 = 2$$

$$y = 2e^{2x} + 2e^{-x} - 2x^2 + 2x - 3$$

$$b) y'' - y' - 2y = e^{3x}$$

$$y_H | \quad y'' - y' - 2y = 0$$

$$\lambda^2 - \lambda - 2 = 0 \Rightarrow \lambda = \frac{1 \pm \sqrt{1+8}}{2} = \frac{1 \pm 3}{2} \rightarrow \begin{matrix} 2 \\ -1 \end{matrix}$$

$$y = c_1 e^{2x} + c_2 e^{-x}$$

$$y_P | \quad y = u e^{2x} + v e^{-x}$$

$$\begin{cases} u' e^{2x} + v' e^{-x} = 0 \\ 2u' e^{2x} - v' e^{-x} = e^{3x} \end{cases}$$

$$W = \begin{vmatrix} e^{2x} & e^{-x} \\ 2e^{2x} & -e^{-x} \end{vmatrix} = -e^{2x-x} - 2e^{2x-x} = -3e^x$$

$$u' = \frac{\begin{vmatrix} 0 & e^{-x} \\ e^{3x} & -e^{-x} \end{vmatrix}}{-3e^x} = \frac{-e^{3x-x}}{-3e^x} = \frac{e^x}{3}$$

$$\frac{du}{dx} = \frac{e^x}{3} \Rightarrow \int du = \int \frac{e^x}{3} dx \Rightarrow u = \frac{e^x}{3}$$

$$v' = \frac{\begin{vmatrix} e^{2x} & 0 \\ 2e^{2x} & e^{3x} \end{vmatrix}}{-3e^x} = \frac{e^{2x+3x}}{-3e^x} = \frac{e^{4x}}{-3}$$

$$\frac{dv}{dx} = \frac{e^{4x}}{-3} \Rightarrow \int dv = \int \frac{e^{4x}}{-3} dx \Rightarrow v = \frac{e^{4x}}{-12}$$

$$y_p = \frac{e^x}{3} e^{2x} + \frac{e^{4x}}{-12} e^{-x} = \frac{e^{3x}}{3} - \frac{e^{3x}}{12} = \frac{e^{3x}}{4}$$

$$y = c_1 e^{2x} + c_2 e^{-x} + \frac{e^{3x}}{4}$$

$$y(0) = 1 \wedge y'(0) = 2$$

$$y(0) = 1 \Rightarrow 1 = c_1 + c_2 + \frac{1}{4} \Rightarrow 1 = c_1 + c_2 + \frac{1}{4} \quad (1)$$

$$y' = 2c_1 e^{2x} - c_2 e^{-x} + \frac{3e^{3x}}{4}$$

$$y'(0) = 2 \Rightarrow 2 = 2c_1 - c_2 + \frac{3}{4} \Rightarrow 2 = 2c_1 - c_2 + \frac{3}{4} \quad (2)$$

$$(1) \wedge (2) \Rightarrow \begin{cases} c_1 = 1 - c_2 - \frac{1}{4} \\ 2 = 2c_1 - c_2 + \frac{3}{4} \end{cases} \Rightarrow \begin{cases} c_1 = \frac{3}{4} - c_2 \\ 2 = 2\left(\frac{3}{4} - c_2\right) - c_2 + \frac{3}{4} \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} c_1 = \frac{3}{4} - c_2 \\ 2 = \frac{3}{2} - 2c_2 - c_2 + \frac{3}{4} \end{cases} \Rightarrow \begin{cases} c_1 = \frac{3}{4} - c_2 \\ 3c_2 = \frac{9}{4} - 2 \end{cases} \Rightarrow \begin{cases} c_2 = \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12} \\ c_1 = \frac{3}{4} - \frac{1}{12} = \frac{2}{3} \end{cases}$$

$$\boxed{y = \frac{2}{3} e^{2x} + \frac{1}{12} e^{-x} + \frac{e^{3x}}{4}}$$

$$c) \quad y'' + 2y' + 2y = \sin 2x + \cos 2x$$

$$y_h \quad y'' + 2y' + 2 = 0$$

$$\lambda^2 + 2\lambda + 2 = 0 \Rightarrow \lambda = \frac{-2 \pm \sqrt{4-8}}{2} = \frac{-2 \pm \sqrt{-4}}{2} \Rightarrow$$

$$\Rightarrow \lambda = -1 \pm i \quad \begin{matrix} \nearrow \lambda = -1 + i \\ \searrow \lambda = -1 - i \end{matrix}$$

$$y = e^{-x} (A \cos x + B \sin x)$$

$$y_p = H \sin 2x + K \cos 2x$$

$$y' = 2H \cos 2x - 2K \sin 2x$$

$$y'' = -4H \sin 2x - 4K \cos 2x$$

$$y'' + 2y' + 2y = \overbrace{-4H \sin 2x - 4K \cos 2x} + \overbrace{4H \cos 2x - 4K \sin 2x} +$$

$$+ \overbrace{2H \sin 2x + 2K \cos 2x} = (-2H - 4K) \sin 2x +$$

$$+ (-2K + 4H) \cos 2x = \sin 2x + \cos 2x \Rightarrow$$

$$\Rightarrow \begin{cases} -2H - 4K = 1 \\ -2K + 4H = 1 \end{cases} \Rightarrow \begin{cases} H = \frac{1+2K}{4} \\ -2\left(\frac{1+2K}{4}\right) - 4K = 1 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} \frac{-1-2k}{2} - 4k = 1 \\ H = \frac{1+2k}{4} \end{cases} \Rightarrow \begin{cases} -\frac{1}{2} - 5k = 1 \\ H = \frac{1+2k}{4} \end{cases} \Rightarrow$$

$$\Rightarrow k = \frac{-\frac{1}{2} - 1}{5} = -\frac{\frac{3}{2}}{5} = -\frac{3}{10}$$

$$H = \frac{1 - \frac{6}{10}}{4} = \frac{\frac{4}{10} \cdot \frac{1}{4}}{1} = \frac{1}{10}$$

$$y_p = \frac{1}{10} \sin 2x - \frac{3}{10} \cos 2x$$

$$y_g = e^{-x} (A \cos x + B \sin x) + \frac{1}{10} \sin 2x - \frac{3}{10} \cos 2x$$

$$\Rightarrow y(0) = 0 \wedge y'(0) = 1$$

$$0 = 1(A) - \frac{3}{10} \Rightarrow A = \frac{3}{10}$$

$$y' = -e^{-x} (A \cos x + B \sin x) + e^{-x} (-A \sin x + B \cos x) + \frac{1}{5} \cos 2x + \frac{3}{5} \sin 2x$$

$$1 = -1(A) + 1(B) + \frac{1}{5} \Rightarrow B = 1 + A - \frac{1}{5} = 0$$

$$\Rightarrow B = 1 + \frac{3}{10} - \frac{2}{10} = \frac{10+3-2}{10} = \frac{11}{10}$$

$$y_p = e^{-x} \left(\frac{3}{10} \cos x + \frac{11}{10} \sin x \right) + \frac{1}{10} \sin 2x -$$

$$- \frac{3}{10} \cos(2x)$$

$$8) \downarrow) y''' - 3y'' + 2y' = x^2 - x$$

$$J_H) r^3 - 3r^2 + 2r = 0$$

$$r(r^2 - 3r + 2) = 0 \Rightarrow r(r-1)(r-2) = 0 \Rightarrow \begin{matrix} r=0 \\ r=1 \\ r=2 \end{matrix}$$

$$y_H = c_1 e^{0x} + c_2 e^x + c_3 e^{2x}$$

$$J_P) y_P = Ax^3 + Bx^2 + Cx + D \quad y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B \quad y'''_P = 6A$$

$$y''' - 3y'' + 2y' = 6A - 18Ax - 6B + 6Ax^2 + 4Bx + 2C = x^2 - x$$

$$\Rightarrow 6Ax^2 + (-18A + 4B)x + (6A - 6B + 2C) = x^2 - x$$

$$6A = 1 \Rightarrow A = 1/6$$

$$-18 \cdot \frac{1}{6} + 4B = -1 \Rightarrow 4B = 2 \Rightarrow B = \frac{1}{2}$$

$$6 \cdot \frac{1}{6} - 6 \cdot \frac{1}{2} + 2C = 0 \Rightarrow 2C = 3 - 1 \Rightarrow C = 1$$

because $D = 0$

$$y_P = \frac{1}{6}x^3 + \frac{1}{2}x^2 + x$$

$$J_G) y_G = c_1 + c_2 e^x + c_3 e^{2x} + \frac{1}{6}x^3 + \frac{1}{2}x^2 + x$$

$$y(0) = 0 \Rightarrow 0 = c_1 + c_2 + c_3$$

$$y'(0) = 1 \Rightarrow 1 = c_2 + 2c_3 + 1$$

$$y''(0) = 2 \Rightarrow 2 = c_2 + 4c_3 + 2 \cdot \frac{3}{6} \cdot 0 + 2 \cdot \frac{1}{2}$$

$$c_1 + c_2 + c_3 = 0$$

$$c_2 + 2c_3 = 1 \Rightarrow c_2 = 1 - 2c_3$$

$$2 = c_2 + 4c_3 + 1 \Rightarrow 1 = 1 - 2c_3 + 4c_3 \Rightarrow 2c_3 = 0 \Rightarrow c_3 = 0$$

Luego $c_2 = 1$

$$c_1 = -c_3 - c_2 \Rightarrow c_1 = -1$$

$$y = -1 + e^x + \frac{1}{6}x^3 + \frac{1}{2}x^2 + x$$

Halle la SP de $y'' - y = 2x - 1$ con recta normal de ecuación $x + 3y = 6$ en $(0, y_0)$

$$8) e) \quad y'' - y = 2x - 1$$

$$y = y_H + y_P$$

$$y_H: y'' - y = 0$$

$$y_H \quad r^2 - 1 = 0 \Rightarrow (r-1)(r+1) = 0 \quad \begin{matrix} r=1 \\ r=-1 \end{matrix}$$

$$y_H = c_1 e^{-x} + c_2 e^x$$

$$y_P = Ax + B \quad y' = A \quad y'' = 0$$

$$0 - Ax - B = 2x - 1 \Rightarrow A = -2 \quad B = 1$$

$$y_P = -2x + 1$$

$$y = c_1 e^{-x} + c_2 e^x - 2x + 1$$

recta normal $x + 3y = 6$

$$y = \frac{6-x}{3} = 2 - \frac{x}{3}$$

$$x=0 \Rightarrow y=2$$

$$x=0 \Rightarrow y' = 3 \quad (\text{pendiente de la recta tge})$$

$$y(0)=2: 2 = c_1 + c_2 + 1$$

$$y'(0)=3: 3 = -c_1 + c_2 - 2$$

$$y = -c_1 e^{-x} + c_2 e^x$$

$$c_1 = 1 - c_2$$

$$3 = -1 + c_2 + c_2 - 2 \Rightarrow c_2 = 3$$

$$6 = 2c_2 \Rightarrow c_2 = 3 \quad c_1 = 1 - 3 = -2 \quad c_1 = -2$$

$$c_1 = -2 \quad c_2 = 3$$

$$y = -2e^{-x} + 3e^x - 2x + 1$$

La pendiente de la recta normal es igual a $-1/3$, por lo cual la pendiente de la recta tangente es igual a 3

$$9) \quad y'' + y = f(x) \quad \left. \begin{array}{l} y = 2x^2 \text{ es SP} \\ y' = 4x \\ y'' = 4 \end{array} \right\} 4 + 2x^2 = f(x)$$

Luego $y'' + y = 4 + 2x^2$

$$y_H \quad y'' + y = 0 \quad \begin{array}{l} k^2 + 1 = 0 \\ \quad \quad \quad \nearrow k = i \\ \quad \quad \quad \searrow k = -i \end{array}$$

$$y_H = e^{0x} (C_1 \cos x + C_2 \operatorname{sen} x) = C_1 \cos x + C_2 \operatorname{sen} x$$

$$y_P \quad \left. \begin{array}{l} y = ax^2 + bx + c \\ y' = 2ax + b \\ y'' = 2a \end{array} \right\} 2a + ax^2 + bx + c = 4 + 2x^2 \Rightarrow$$

$$\Rightarrow a = 2 \quad b = 0 \quad \begin{array}{l} 2a + c = 4 \\ c = 4 - 4 = 0 \end{array}$$

$$y_P = 2x^2$$

$$y_G = C_1 \cos x + C_2 \operatorname{sen} x + 2x^2$$

$$10) Mx'' + dx' + kx = F$$

$$a) 3x'' + 12x' + 87x = 4,35$$

$$x_H \mid 3k^2 + 12k + 87 = 0$$

$$k = \frac{-12 \pm \sqrt{12^2 - 4 \cdot 3 \cdot 87}}{6} \rightarrow k_1 = \frac{-12 + 30i}{6} = -2 + 5i$$

$$k_2 = \frac{-12 - 30i}{6} = -2 - 5i$$

$$x_H = e^{-2t} (C_1 \cos(5t) + C_2 \sin(5t))$$

$$x_p \mid \begin{array}{l} x = a \\ x' = 0 \\ x'' = 0 \end{array} \quad \begin{array}{l} 3 \cdot 0 + 12 \cdot 0 + 87 \cdot a = 4,35 \\ a = \frac{4,35}{87} = 0,05 \end{array}$$

$$X = e^{-2t} [C_1 \cos(5t) + C_2 \sin(5t)] + 0,05$$

$$X(0) = 0 \quad C_1 + 0,05 = 0 \Rightarrow C_1 = -0,05$$

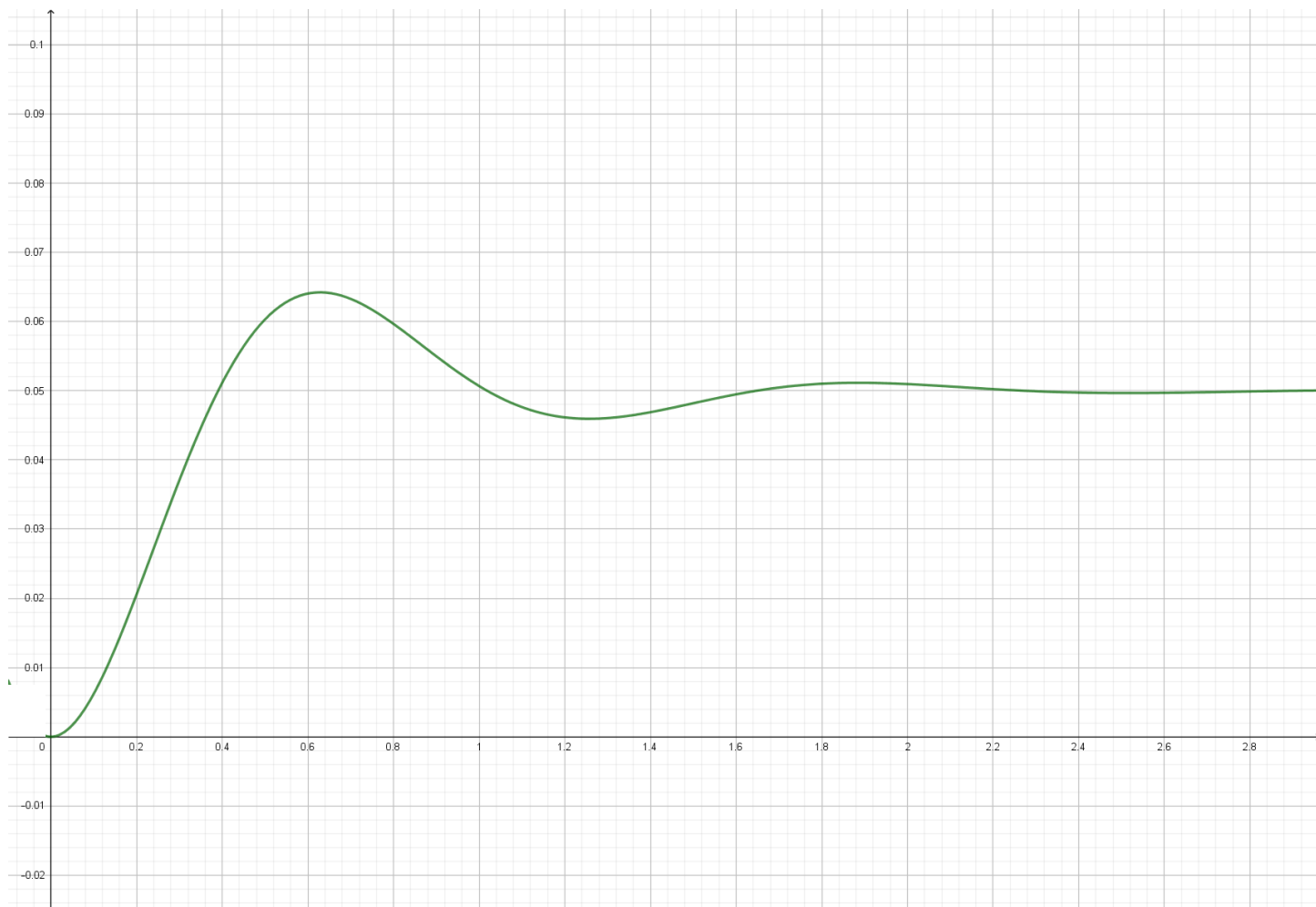
$$X'(0) = 0 \quad X' = -2e^{-2t} (C_1 \cos(5t) + C_2 \sin(5t)) + e^{-2t} [-5C_1 \sin(5t) + 5C_2 \cos(5t)]$$

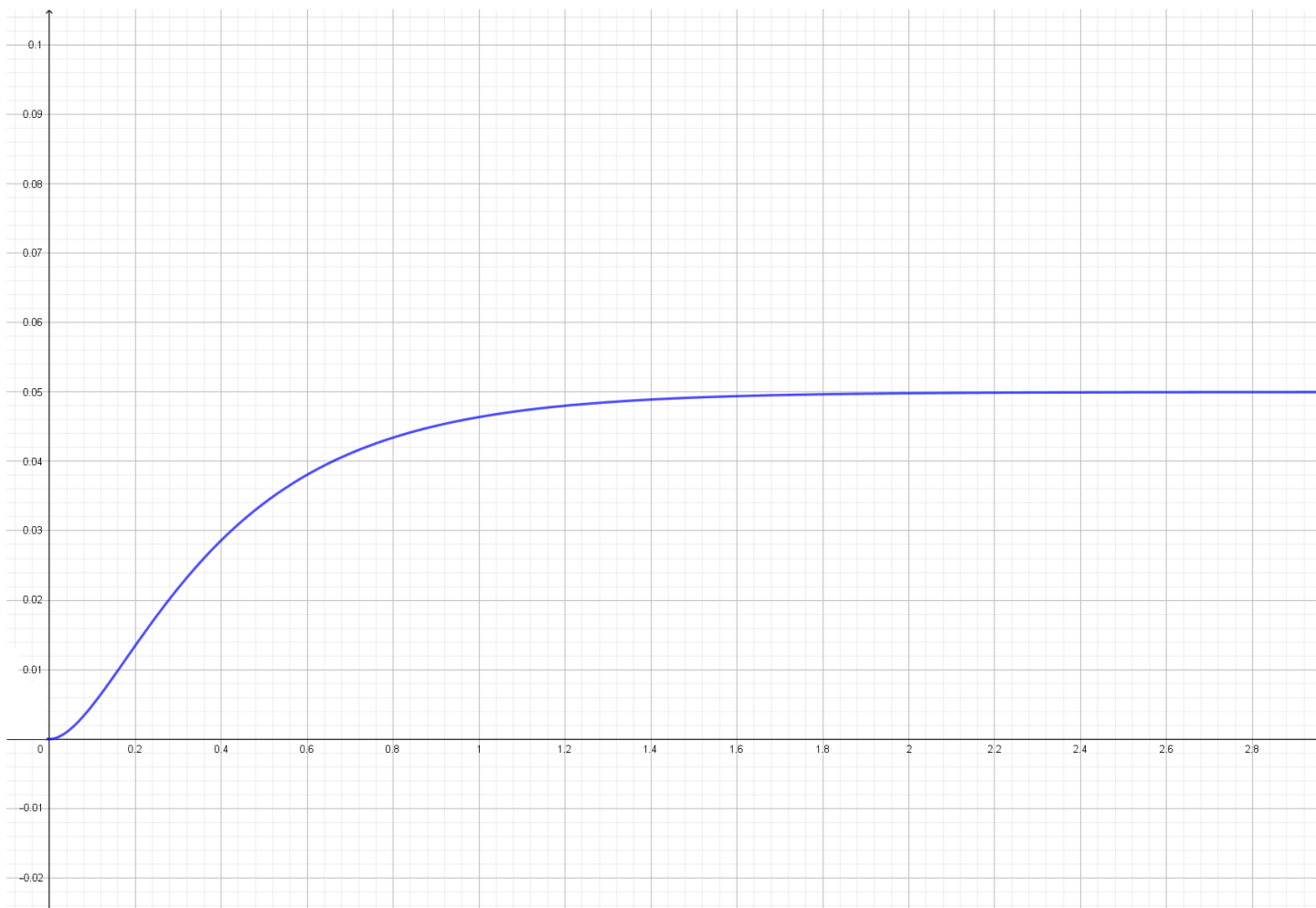
$$-2(C_1) + (5C_2) = 0$$

$$5C_2 = 2C_1 = 2(-0,05)$$

$$C_2 = \frac{2(-0,05)}{5} = -0,02$$

$$x(t) = e^{-2t} (-0,05 \cos(5t) - 0,02 \sin(5t)) + 0,05$$





10)b) $[x'] = \frac{m}{\text{seg}}$ $[x''] = \frac{m}{\text{seg}^2}$

$[M] = \text{kg}$ $M \cdot x'' \rightarrow \text{unidad de fuerza} \left[N = \frac{\text{kg} \cdot m}{\text{seg}^2} \right]$

$[k] = \frac{\text{kg}}{\text{seg}^2} = \frac{N}{m}$ $k \cdot x \rightarrow \text{unidad de fuerza} \left[N = \frac{\text{kg}}{\text{seg}^2} \cdot m \right]$

$[d] = \frac{k}{\text{seg}} = \frac{N \cdot \text{seg}}{m}$ $d \cdot x' \rightarrow \text{unidad de fuerza} \left[N = \frac{k}{\text{seg}} \cdot \frac{m}{\text{seg}} \right]$

$$11) \quad \frac{1}{29} r'' + \frac{4}{29} r' + r = 0,05$$

$$\left. \begin{aligned} \frac{R}{L} = 4 &\Rightarrow R = 4L \\ \frac{1}{LC} = 29 &\Rightarrow L = \frac{1}{29C} \end{aligned} \right\} \Rightarrow R = 4 \cdot \frac{1}{29C} \Rightarrow R_C = \frac{4}{29}$$

$$r_H \quad \frac{1}{29} k^2 + \frac{4}{29} k + 1 = 0$$

$$k_{1,2} = \frac{-\frac{4}{29} \pm \sqrt{\frac{16}{29^2} - 4 \cdot \frac{1}{29}}}{\frac{2}{29}} = -2 \pm \sqrt{\frac{-100}{841}} =$$

$$= -2 \pm \frac{10i}{2} \rightarrow -2 + 5i$$

$$\rightarrow -2 - 5i$$

$$r_H = e^{-2t} (c_1 \cos(5t) + c_2 \sin(5t))$$

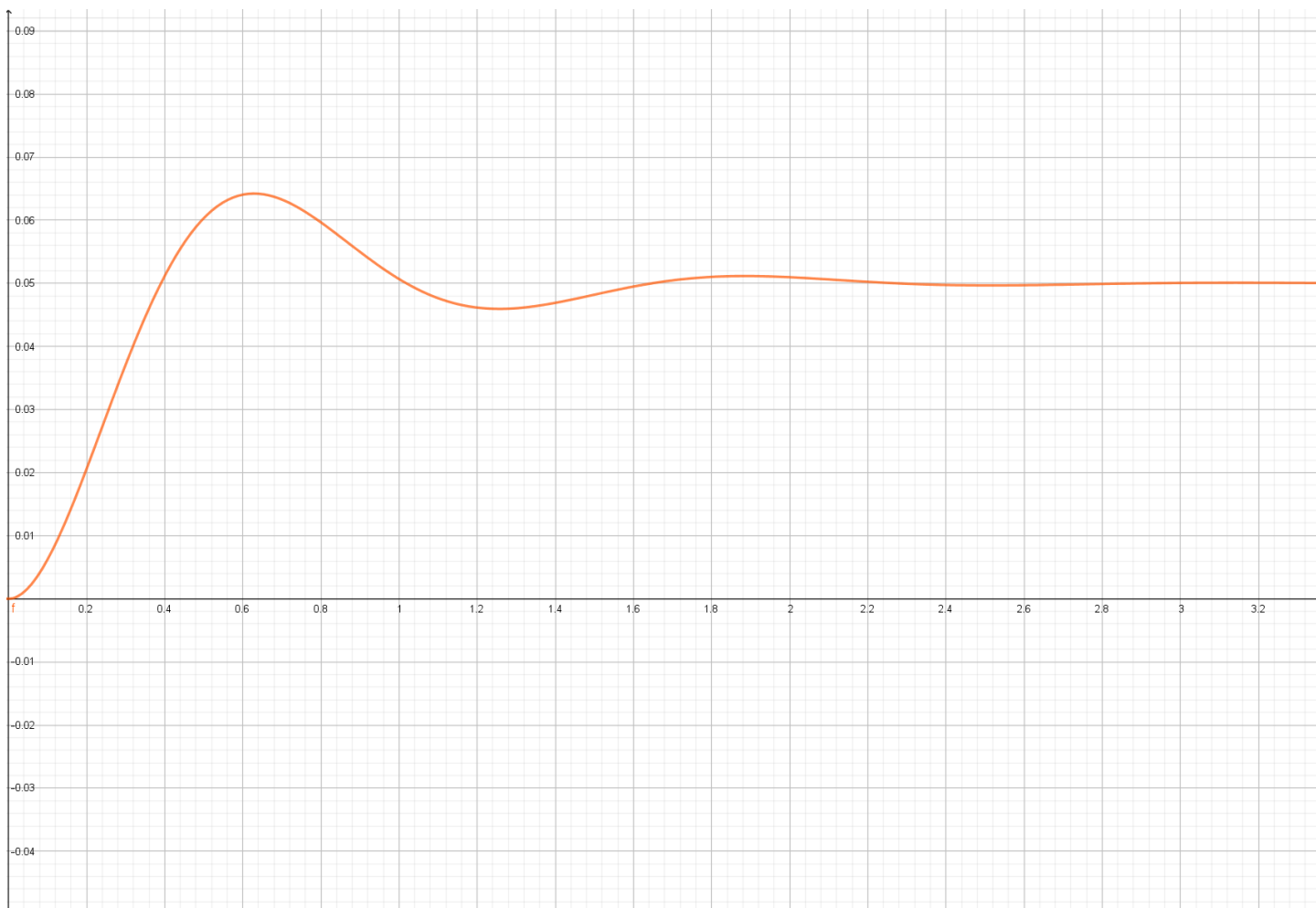
$$r_P \quad \begin{aligned} r &= a \\ r' &= 0 \\ r'' &= 0 \end{aligned} \quad \frac{1}{29} \cdot 0 + \frac{4}{29} \cdot 0 + a = 0,05 \Rightarrow a = 0,05$$

$$r(t) = e^{-2t} (c_1 \cos(5t) + c_2 \sin(5t)) + 0,05$$

$$r(0) = 0 \Rightarrow c_1 + 0,05 = 0 \Rightarrow c_1 = -0,05$$

$$\begin{aligned} r'(0) = 0 &\Rightarrow -2(c_1 \cdot 1 + c_2 \cdot 0) + (-5c_1 \cdot 0 + 5c_2) = 0 \Rightarrow \\ &\Rightarrow -2 \cdot (-0,05) + 5c_2 = 0 \Rightarrow c_2 = -\frac{0,1}{5} = -0,02 \end{aligned}$$

$$r(t) = e^{-2t} (-0,05 \cos(5t) - 0,02 \sin(5t)) + 0,05$$



$$12) r'' + kr + 9r = 0$$

$$z^2 + kz + 9 = 0 \Rightarrow z_{1,2} = \frac{-k \pm \sqrt{k^2 - 36}}{2}$$

Oscilación amortiguada: raíces complejas conjugadas

$$k^2 - 36 < 0 \Rightarrow k^2 < 36 \Rightarrow |k| < 6 \Rightarrow \boxed{-6 < k < 6} \quad (1)$$

$k = RC \rightarrow$ capacitancia
 $\downarrow$
 resistencia

son valores siempre positivos. Luego $\boxed{k > 0} \quad (2)$

Por (1) y (2): $\boxed{0 < k < 6}$

$$13) a) \quad f(x, y) = (2y - x, x)$$

$$y' = \frac{Q(x, y)}{P(x, y)} \Rightarrow y' = \frac{x}{2y - x} \quad \text{homog. 1° order}$$

$$z = y/x \Rightarrow y = zx \Rightarrow y' = z'x + z$$

$$z'x + z = \frac{x}{2zx - x} = \frac{1}{2z - 1}$$

$$\frac{dz}{dx} x = \frac{1}{2z - 1} - z = \frac{1 - 2z^2 + z}{2z - 1}$$

$$\int dz \frac{2z - 1}{-2z^2 + z + 1} = \int \frac{dx}{x}$$

$$-\frac{2}{3} \ln(2z + 1) - \frac{1}{3} \ln(z - 1) = \ln(kx)$$

$$\ln(2z + 1)^{-2/3} - \ln(z - 1)^{1/3} = \ln(kx)$$

$$\ln \left[\frac{(2z + 1)^{-2/3}}{(z - 1)^{1/3}} \right] = \ln(kx) = 0$$

$$= 0 \quad \frac{1}{(2z + 1)^{2/3} (z - 1)^{1/3}} = kx$$

$$= 0 \quad \frac{1}{\left(2 \frac{y}{x} + 1\right)^{2/3} \left(\frac{y}{x} - 1\right)^{1/3}} = kx$$

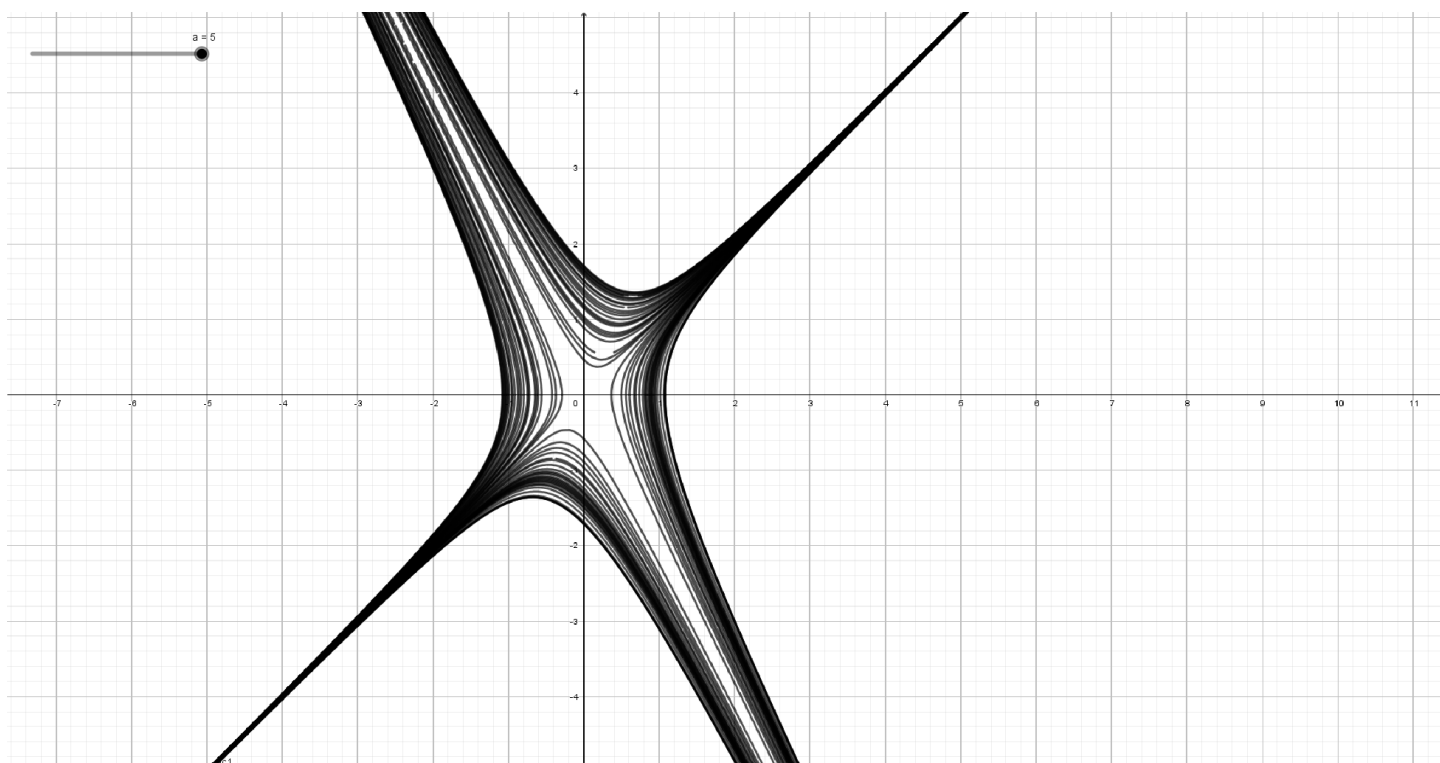
$$\frac{1}{x} = k \left(\frac{2y + x}{x} \right)^{2/3} \left(\frac{y - x}{x} \right)^{1/3}$$

$$\frac{1}{x} = k \frac{(2y+x)^{2/3}}{x^{2/3}} \frac{(y-x)^{1/3}}{x^{1/3}} \Rightarrow$$

$$\Rightarrow 1 = k \sqrt[3]{(2y+x)^2 (y-x)} \Rightarrow$$

$$\Rightarrow 1 = C (2y+x)^2 (y-x) = -C (2y+x)^2 (x-y) \Rightarrow$$

$$\Rightarrow \boxed{a = (2y+x)^2 (x-y)}$$

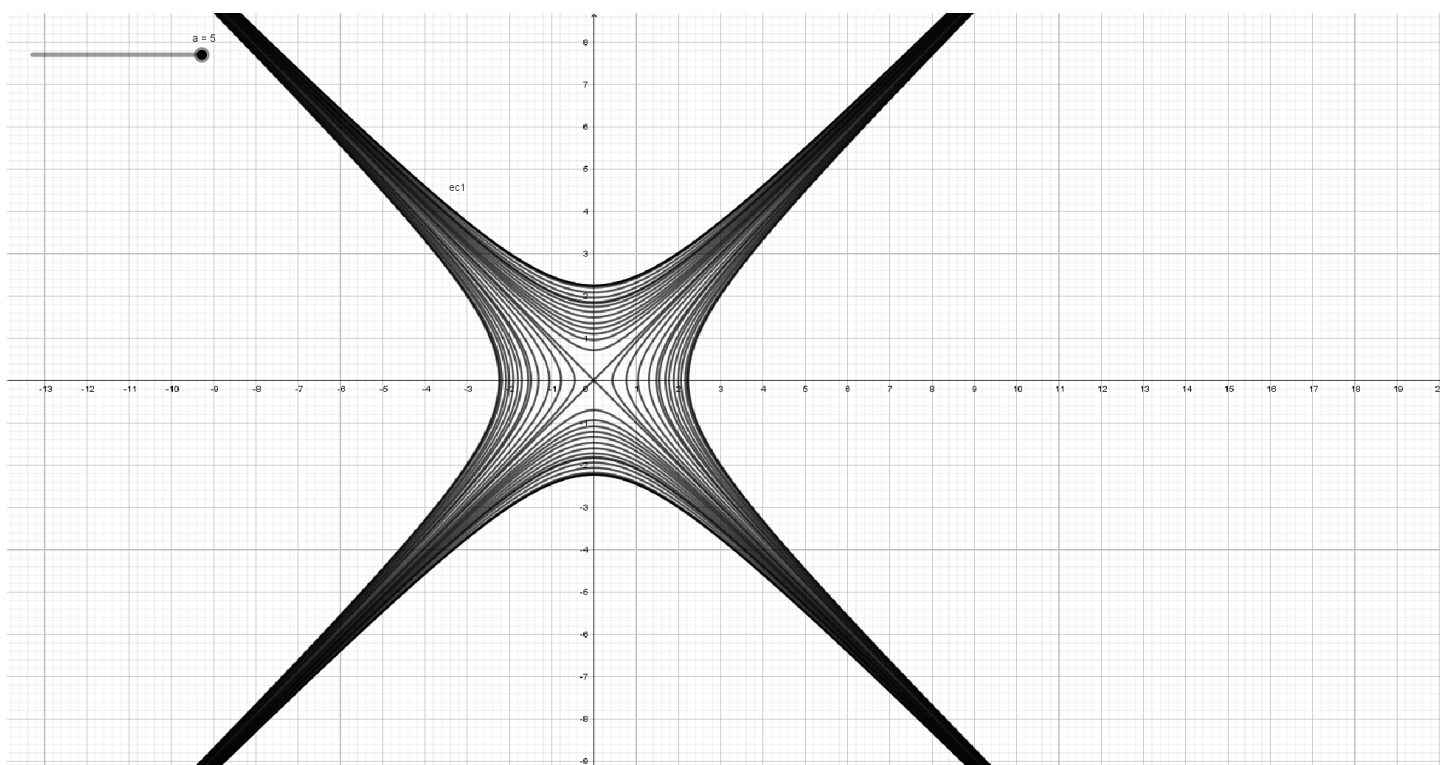


$$13)b) \quad \vec{f}(x,y) = (y, x)$$

$$y' = \frac{x}{y} \Rightarrow \frac{dy}{dx} = \frac{x}{y} \Rightarrow \int dy \, y = \int x \, dx \Rightarrow$$

$$\Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow \frac{y^2 - x^2}{2} = C \Rightarrow$$

$$\Rightarrow \boxed{y^2 - x^2 = k}$$

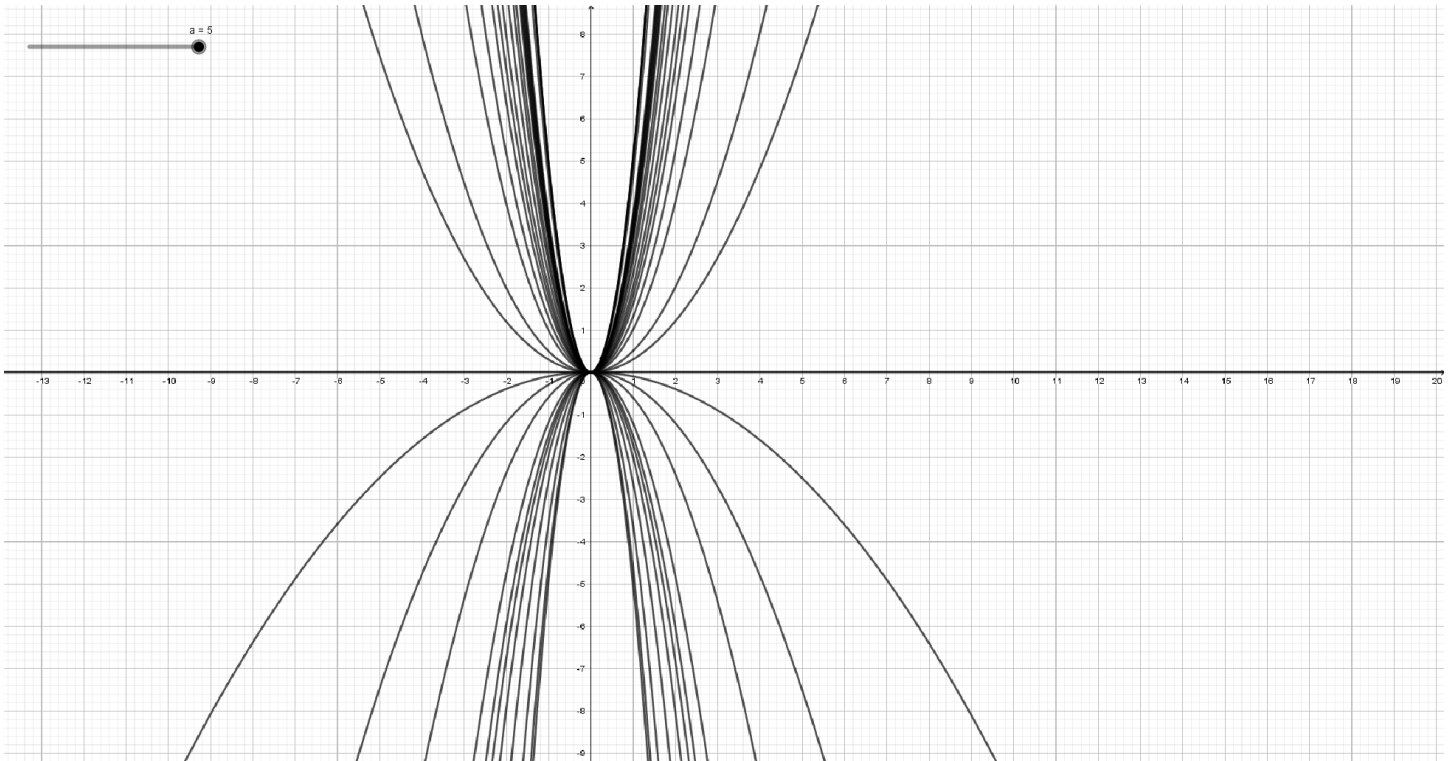


$$13)c) \vec{f}(x,y) = \left(\frac{x}{2}, y \right)$$

$$y' = \frac{y}{x/2} \Rightarrow \frac{dy}{dx} = \frac{2y}{x} \Rightarrow \int \frac{dy}{y} = \int \frac{2}{x} dx \Rightarrow$$

$$\Rightarrow \ln(ky) = 2 \ln x = \ln x^2 \Rightarrow$$

$$\Rightarrow ky = x^2 \Rightarrow \boxed{y = c x^2}$$



13)d) $\vec{f}(x,y) = (kxy^2, x^2y)$ k constante

$$y' = \frac{x^2y}{kxy^2} \Rightarrow \frac{dy}{dx} = \frac{x}{ky} \Rightarrow \int dy \, ky = \int dx \, x \Rightarrow$$

$$\Rightarrow \frac{ky^2}{2} = \frac{x^2}{2} + C \Rightarrow \frac{ky^2}{2} - \frac{x^2}{2} = C \Rightarrow$$

$$\Rightarrow \boxed{ky^2 - x^2 = a}$$

Gráfico realizado para $k=1$

